FOR TRIAL PURPOSE ONLY

MATERIAL SAFETY DATA SHEET (MSDS)
(HP FurnOKare)

(HP FurnOKare)

Section 1. Chemical Product and Company Identification

Product Name	HP FurnOKare
Product No.	HP FurnOKare
Application	Furnace cleaning
Supplier	HPCL R&D Centre, Bangalore
Emergency Telephone	080-28078650

Section 2. Hazards Identification

Hazard Identification

Hazardous in case of inhalation (lung damage). Hazardous in case of skin contact (itching), eye exposure (irritation) and respiratory system (irritation to mucous membrane). Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

White and *Blue crystals; slight odor.*
WARNING! Strong oxidizing agent and moderately toxic by ingestion
Target organs: Liver, Kidney, Central Nervous System

Classification as per Regulation (EC) No 1272/2008
Oxidising solid, Category 3, H272 –May intensify fire; oxidizer.
H315 –May cause skin irritation.

H319 –May cause eye irritation.
Precautionary Statement P220- Keep store away from clothing/combustible material

Safety Phrases
Use personal protection equipment and ammonia risk methods. Thoroughly rinse with plenty of water in case of contact with eyes and seek medical advice. Wear suitable gloves. Use only in well-ventilated areas. Kindly avoid release to the environment. Refer to special instructions/safety data sheets. Seek medical advice immediately and show this sheet if swallowed.

DG Class: Class 3: Flammable and explosive Solids

<u>Section 3. First Aid Measures</u>

Always seek professional medical attention after first aid measures are provided.
Eyes: Immediately flush eyes with excess water for 15 minutes, lifting lower and upper eyelids occasionally.
Skin: Immediately flush skin with excess water for 15 minutes while removing contaminated clothing.
Ingestion:
Do NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Loosen tight clothing such as a collar, tie, belt or waistband. Get medical attention if symptoms appear.Induce vomiting immediately.
Ulcerations of the mouth, vomiting of blood, and rapid appearance of shock, convulsions, twitching,
tetany, and cardiovascular collapse may occur following ingestion of oxalic acid or its soluble salts.
Systemic effects may be due to formation of calcium oxalate which is insoluble at physiological pH
and can be deposited in the brain and kidney tubules. Resultant hypocalcemia might disturb the
Inhalation: Remove to fresh air. If not breathing, give artificial respiration.

<u>Section 4. Composition and Information on Ingredients</u>

Name	CAS No:	Content
Mixed metal nitrates	7631-99-4	0.01 to 60 %
Organic surfactant mixtures	57-13-6	0.01 to 30 %

Section 5. Stability and Reactivity

Avoid heat and ignition sources.
Stability: Stable under normal conditions of use and storage.
Incompatibility: Strong oxidizing agents
Shelf life: Poor shelf life, store in cool, dry environment

Stability Not available (NA).
Instability Temperature NA
Hazardous Decomposition Products
Vapours/gases/fumes of: Carbon dioxide (CO2), ammonia (NH3) Oxides of: nitrogen and carbon, formic acid.

Materials to Avoid
Strong oxidising substances
Conditions to Avoid
Avoid contact with strong oxidisers.

Chemical Stability:
Stable under normal temperatures and pressures.
Conditions to Avoid:
Dust generation, excess heat. Oxalates slowly corrode steel.
Incompatibilities with Other Materials:
Strong oxidizing agents.

Section 6. Physical and Chemical Properties

Appearance: Form, solid, crystalline powder
Colour: White
Odour: Odourless
Taste: Bitter saline
pH (1% solution): 6.5 – 8.5
Boiling point: Not Available
Melting Point: 222°C range
Specific gravity: Density 2.1 (Water=1)
Bulk Density: 70-80 (lb/ft3)
Specific Heat: 0.29 (BTU/lb°F)
Solubility: Easily soluble in cold and hot water

Water solubility: 320 g/l water at 20°C
Odour threshold: Not applicable
Flash point: Does not flash
Evaporation rate: No information available
Flammability: The product is not flammable
Explosive properties: Not classified as explosive
Oxidising properties: Oxidiser
Other information: None

Section 7. Fire Fighting Measures

EXTINGUISHING MEDIA
Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide. Use water spray to cool fire-exposed containers. Use water only!

SPECIFIC HAZARDS DURING FIRE FIGHTING
Not combustible.
Has a fire-promoting effect due to release of oxygen.
Ambient fire may liberate hazardous vapours.
Fire may cause evolution of: nitrogen oxides

SPECIAL PROTECTIVE EQUIPMENT FOR FIRE FIGHTERS
Do not stay in dangerous zone without self-contained breathing apparatus. In order to avoid contact with skin, keep a safety distance and wear fully protective impervious suit.

FURTHER INFORMATION
Suppress (knock down) gases/Vapour/mists with a water spray jet. Prevent fire extinguishing water from contaminating surface water or the ground water system. Use water spray to cool unopened containers.

Section 8. Toxicology Information

Acute Symptoms/Signs of exposure: ***Eyes***: Redness, tearing, itching, burning, conjunctivitis.
Skin: Redness, itching.
Ingestion: Irritation and burning sensations of mouth and throat, nausea, vomiting and abdominal pain. ***Inhalation***: Irritation of mucous membranes, coughing, wheezing, shortness of breath,
Chronic Effects: No information found.
Sensitization: none expected
Acute oral toxicity: LD50 Oral (rat)->2000mg/kg
Acute dermal toxicity: LD50 (rat)- >5000 mg/kg
Acute inhalation toxicity: LC50 (rat) > 0.527 (4 h)
Irritation of the eyes: Causes irritation
Irritation of the skin: Causes irritation
RTECS#: TT3700000
Carcinogenicity: Not listed by ACGIH, IARC, EU, NTP or OSHA

Potential health effects: Inhalation: May be harmful if inhaled. May cause respiratory tract irritation.
Ingestion: Toxic if swallowed.
Skin: May be harmful if absorbed through skin. May cause skin irritation.
Eyes: May cause eye irritation.

Section 9. Ecological information

Ecotoxicity
Acute aquatic toxicity: LC50 96 h fresh water fish : 1378 mg/L
Fish: LC50 48h Daphnia magna: 490 mg/L
Biodegradability: In principle only abiotic degradation processes are relevant for the substance. In aqueous solutions, the substance will dissociate into Potassium and nitrate ions. Under anoxic conditions, denitrification occurs and nitrate is ultimately converted into nitrogen as a part of nitrogen cycle.
KNO3 has low potential for bioaccumulation based on Physical chemical properties (High water solubility)
Additional ecological information
Nitrate has a low potential for adsorption. Portion not taken up by plants, can leach to ground water.
Other adverse effects

Section 10. Exposure Controls / Personal Protection

Use ventilation to keep airborne concentrations below exposure limits. Have approved eyewash facility, safety shower, and fire extinguishers readily available. Wear chemical splash goggles and chemical resistant clothing such as gloves and aprons. Wash hands thoroughly after handling material and before eating or drinking.

Section 11. Accidental Release Measures

PERSONAL PRECAUTIONS
Use personal protective equipment. Avoid dust formation. Avoid breathing dust. Ensure adequate ventilation. Evacuate personnel to safe areas.
ENVIRONMENTAL PRECAUTIONS
Prevent further leakage or spillage if safe to do so. Do not let product enter drains. Discharge into the environment must be avoided.
SPILL CLEAN UP METHODS
Pick up and arrange disposal without creating dust. Keep in suitable, closed containers for disposal.

Section 12. Handling and Storage

Handling: Use with adequate ventilation and do not breathe dust or vapor. Avoid contact with skin, eyes, or clothing. Wash hands thoroughly after handling.

Storage: Store in Oxidizer Storage Area [Yellow Storage] with other oxidizers and away from any combustible materials. Store in a cool, dry, well-ventilated, locked store room away from incompatible materials.

USAGE PRECAUTIONS
Avoid formation of dust and aerosols.
Provide appropriate exhaust ventilation at places where dust is formed. Keep away from sources of ignition - No smoking. Keep away from combustible material.
STORAGE CONDITIONS
Store in a cool place. Keep container tightly closed in a dry and well-ventilated place. Do not store near combustible materials and reducing agents. Hygroscopic.

Exposure controls / Personal protection
ENGINEERING MEASURES
Provide adequate ventilation, including appropriate local extraction, to ensure that the defined workplace exposure limit (WEL) is not exceeded.
PERSONAL PROTECTIVE EQUIPMENT
Protective clothing should be selected specifically for the working place, depending on concentration and quality of the hazardous substances handled. The resistance of the protective clothing to chemicals should be ascertained with the respective supplier.
RESPIRATORY PROTECTION
Where risk assessment shows air-purifying respirators are appropriate use a full-face particle respirator type N100 (US) or type P3 (EN 143) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).
HAND PROTECTION
The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it. Handle with gloves.
EYE PROTECTION
Face shield and safety glasses
SKIN AND BODY PROTECTION
Choose body protection according to the amount and concentration of the dangerous substance at the work place.
HYGIENE MEASURES
Avoid contact with skin, eyes and clothing. Wash hands before breaks and immediately after handling the product. Immediately change contaminated clothing. Apply skin protective barrier cream. Use adequate ventilation to keep airborne concentrations low.

Section 13. Disposal considerations

PRODUCT
Contact a licensed professional waste disposal service to dispose of this material. Dissolve or mix the material with a combustible solvent and burn in a chemical incinerator equipped with an afterburner and scrubber. Observe all federal, state, and local environmental regulations.
CONTAMINATED PACKAGING

Empty container should be taken for local recycling, recovery or waste disposal.

Section 14. Transport Information

Land Transport (ADR/RID)
UN Number: UN 1479
Proper Shipping Name: OXIDISING SOLID, N.O.S.
Class: 5.1
Packing group: III
Environmentally hazardous: --
Air Transport (IATA)
UN Number: UN 1479
Proper Shipping Name: OXIDISING SOLID, N.O.S.
Class: 5.1
Packing group: III
Environmentally hazardous: --
Special precaution for user: yes
Packing instruction, passenger: 516, Maximum quantity 25 Kg
Packing instruction, cargo: 518, Maximum quantity 100 Kg
Sea Transport (IMDG)
UN Number: UN 1479
Proper Shipping Name: OXIDISING SOLID, N.O.S.
Class: 5.1
Packing group: III
Environmentally hazardous: --
Marin Pollutant No

Section 15. Other Information

Other special considerations: Not available.
The information in the MSDS is based on data which is considered to be accurate. However, we makes no guarantees or warranty, either expressed or implied on the accuracy or completeness of this information. The conditions of handling, storage, use and disposal are beyond our control and may be beyond our knowledge. For this and other reasons, we do not assume responsibility and expressly disclaim liability for loss, damage or expense arising out of or in any way connected with the handling, storage, use or disposal of this product. This MSDS was prepared and is to be used for this product. If the product is used as a component in another product, this MSDS information may not be applicable.